

EE5240: Electrical Machines Analysis and Control

Assignment-2

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Abstract—This report presents the analysis of two control systems: 1. Cascaded Control of a DC Motor, and 2. Induction Machine Flux Control. In the DC motor control section, the system was first simulated without any speed controller, with a step input of 40A applied for two current loop bandwidths 500Hz and 2500Hz. The system behavior is similar in both cases with the speed increasing unregulated until the maximum converter output voltage is reached. The key difference is that, for the 2500Hz case, the system is marginally stable and exhibits an oscillatory behavior.

Subsequently, a speed controller for two different phase margins 36.9° and 55° , and the corresponding speed, current response, and the disturbance rejection characteristics were obtained. The 55° system shows fewer oscillations, and faster settling, though with a slower rise time compared to the 36.9° system. The former system also shows poor disturbance rejection with its fall time and undershoot being more than the 36.9° system.

In the induction machine control section, the flux response, and disturbance rejection characteristics were obtained for two different control structures: 1. single-loop control, and 2. two-loop control and their performance parameters were recorded. The two-loop control shows superior performance across all parameters compared to the single-loop control.

I. INTRODUCTION

This report discusses the control system design for two different cases:

- 1) Cascaded control of a 10 HP full bridge converter fed separately excited DC motor is analyzed. The system is first simulated without a speed controller, with a step input of 40 A for two current loop bandwidths 500 Hz and 2500 Hz, and the differences in performance are observed. Subsequently, a speed controller is designed using two different methods, and a similar analysis is performed.
- 2) Flux control of a 30kW induction machine for two different loops is analyzed. For each case, the flux transient response, and disturbance rejection characteristics were obtained and the performance of the two is compared.

II. DC MOTOR CASCADED CONTROL

A. Model Parameters

The cascaded control system is designed for a separately excited DC motor with the parameters given in Table I.

TABLE I
DC MACHINE PARAMETERS

Quantity	Symbol	Value
Rated Power	P	10 HP
Armature Rated Voltage	V_a	240 V
Armature Resistance	R_a	1.13 Ω
Armature Inductance	L_a	12.16 mH
Field Rated Voltage	V_f	300 V
Field Resistance	R_f	180 Ω
Field-Armature Mutual Inductance	L_{af}	0.6456 H
Rated Speed	N	1750 rpm
Moment of Inertia	J	0.0425 kg $\cdot$ m ²
Viscous Friction Coefficient	B	0.0034 N $\cdot$ m $\cdot$ s
Motor Constant	K_b	1.076 N $\cdot$ m/A

The power converter used is a four-quadrant DC-DC converter with the following parameters:

TABLE II
DC-DC CONVERTER PARAMETERS

Quantity	Symbol	Value
DC Bus Voltage	V_{dc}	300 V
PWM Carrier Frequency	f_{sw}	10 kHz

B. Simulation Block Diagram

The analysis is carried out in MATLAB/Simulink, and the general block diagram used is shown in Fig. 1.

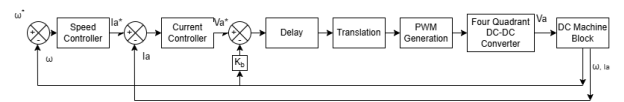


Fig. 1. General Block Diagram of Cascaded Control

1) *Controllers*: These are typical PI controllers whose transfer function is given by eq. 1.

$$G_c = K_p + \frac{K_i}{s} \quad (1)$$

2) *Delay*: Digital implementation of the controllers and sampled PWM usually results in a transport delay of $T_d = 1.5 \frac{1}{f_{sw}}$. The delay is modeled by eq. 2

$$G_d = e^{-sT_d} \quad (2)$$

3) *Power Converter*: The translation block converts the output of the current controller into the required duty cycle, and is given by eq. 3.

$$d^* = \frac{V_a^*}{V_{dc}} \quad (3)$$

The output of the converter is given by eq. 4 and is a typical full bridge DC-DC converter, for the sake of this simulation we have considered ideal switches.

$$V_0 = V_a = (d_1 - d_2)V_{dc} \quad (4)$$

The PWM Generation block utilizes unipolar switching to produce the firing pulses. Each leg of the converter requires its own duty cycle. The output duty ratio from the translation block is divided by two and then added to and subtracted from 0.5. These values are used to generate the gate pulses for both legs of the converter.

4) *DC Machine Block*: The DC machine model available in the *Specialized Power Systems* library of Simscape is used.

C. Current Loop Response for 40A Step Input

The model is run under half rated conditions, when a step input of 40 A is given as a reference to the current loop for two values of loop bandwidths and its response is observed.

1) Current Loop Bandwidth, $f_{bi} = 500 \text{ Hz}$:

The calculated controller parameters are given in the Table III

TABLE III
PI CONTROLLER PARAMETERS

Quantity	Equation	Value
K_{pi}	$2\pi f_{bi} L_a$	38.2018
K_{ii}	$2\pi f_{bi} R_a$	3550
Limits	V_a	$\pm 240 \text{ V}$

From the figures obtained, Fig. 2 to Fig. 4, it can be concluded that, since there is no speed control present, the rotor speed remains unregulated. The current PI controller allows the armature current to track the reference of 40 A, accompanied by a continuous increase in speed.

This increase in speed raises the back emf and, accordingly, the duty cycle input to the converter adjusts so that the converter output voltage rises (Fig. 4). When the output voltage of the converter reaches its maximum value (which is 300 V in this case), the speed stabilizes at 260 rad/s and the armature

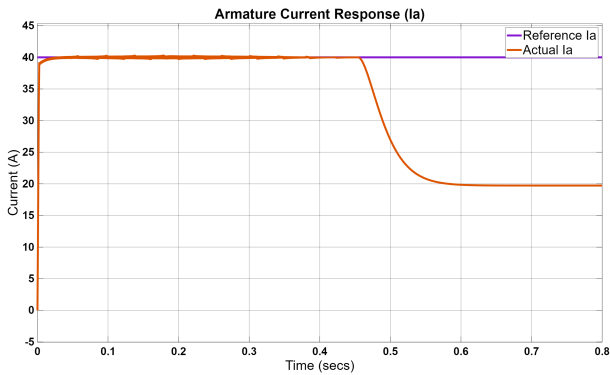


Fig. 2. I_a v/s t

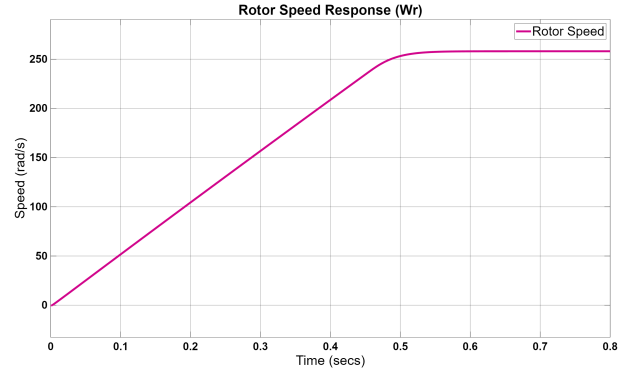


Fig. 3. ω_r v/s t

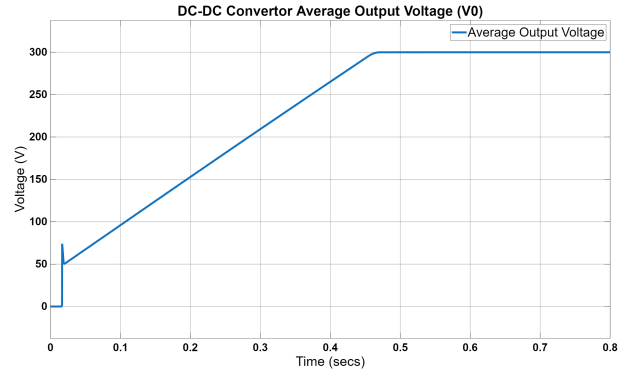


Fig. 4. V_0 v/s t

current ceases to follow the reference and falls to 18.9159 A, calculated from,

$$I_a = \frac{T_l}{K_b} = \frac{20.3535}{1.076}$$

2) Current Loop Bandwidth, $f_{bi} = 2.5 \text{ kHz}$:

The calculated controller parameters are given in Table IV.

TABLE IV
PI CONTROLLER PARAMETERS

Quantity	Equation	Value
K_{pi}	$2\pi f_{bi} L_a$	191.0088
K_{ii}	$2\pi f_{bi} R_a$	17749.9985
Limits	V_a	$\pm 240 \text{ V}$

From the figures obtained, Fig. 5 to Fig. 7, a similar response in the current, speed, and voltage waveforms is observed, with the current tracking the reference, while the speed increases, until the output voltage of the converter reaches its maximum value. Beyond this point, the rotor speed remains constant and the current drops to 18.9126 A. There response of the armature current is oscillatory, indicating that the system is marginally stable compared to the previous case as shown in Fig. 8.

The frequency of oscillations observed from the figure is 1666.667 Hz. The open loop transfer function of the current

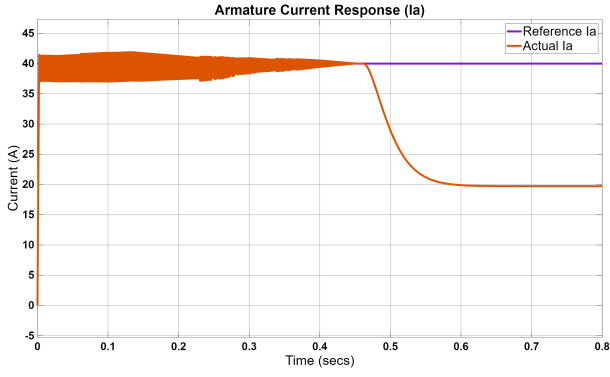


Fig. 5. I_a v/s t

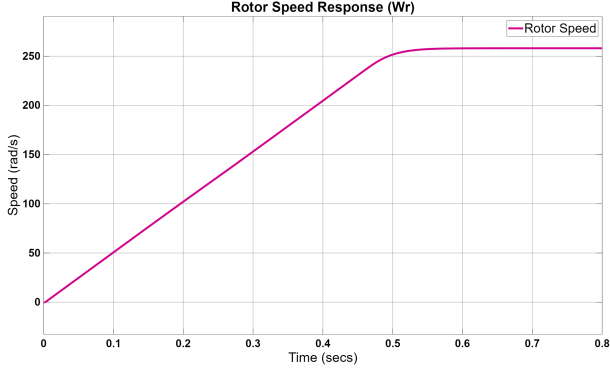


Fig. 6. ω_r v/s t

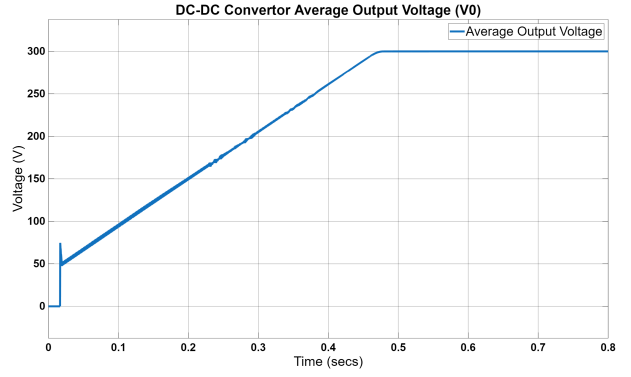


Fig. 7. V_0 v/s t

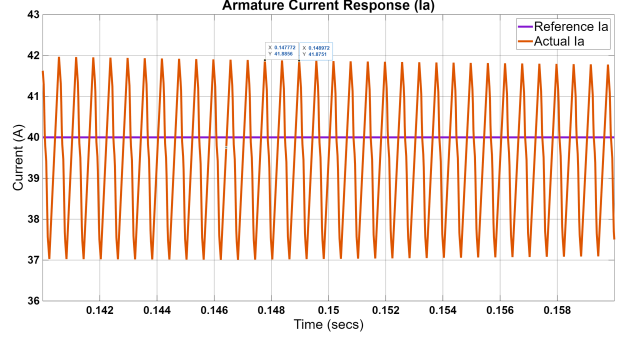


Fig. 8. Oscillations in I_a

control loop with delay is given as,

$$G_0(s) = G_{controller} \cdot G_{delay} \cdot G_{plant}$$

$$G_0(s) = \frac{K_{ii} + sK_{pi}}{s} \frac{1}{R_a + sL_a} e^{-sT_d}$$

Under marginal stable conditions, the phase margin of the system is zero.

$$\angle G_0 = -\pi$$

$$-\frac{\pi}{2} + \arctan \frac{K_{pi}}{K_{ii}} \omega - \omega T_d - \arctan \frac{L_a}{R_a} \omega = -\pi$$

$$\omega T_d = \frac{\pi}{2}$$

$$\omega = \frac{\pi}{2T_d} = 10471.97551 \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = 1666.667 \text{ Hz}$$

For any frequency $f > f_{marginal}$, the system will move to marginal stability.

From Table V, both systems show a similar transient response with the only difference being that the system is marginally stable for 2500 Hz.

TABLE V
COMPARISON BETWEEN $f_{bi} = 500 \text{ Hz}$ AND $f_{bi} = 2500 \text{ Hz}$

f_{bi}	500 Hz	f_{bi}	2500 Hz
t_r	1.8 ms	t_r	1.8 ms
t_s (2%)	4.2 ms	t_s (2%)	—

D. Speed Loop Design

The outer speed loop of the system is designed for two phase margins using the symmetric optimum technique, and the speed, current response and disturbance rejection for a unit step input are observed for both cases. Their corresponding bode plots are also obtained.

1) *Standard $4\tau_{bi}$ Method ($a = 2$):*

The open loop transfer function is given by,

$$G_0(s) = \frac{4\tau_{bi}s + 1}{8\tau_{bi}^3s^3 + 8\tau_{bi}^2s^2}$$

The calculated controller parameters are given in Table VI:

The bode plot of the open loop gain is given in Fig. 9.

It can be observed from Fig. 9 that the maximum phase margin of the system, 36.9° , occurs at the geometric mean of the two corner frequencies of the open loop.

$$\text{Geometric Mean} = \sqrt{\frac{1}{4\tau_{bi}} \frac{1}{\tau_{bi}}} = \frac{1}{2\tau_{bi}}$$

TABLE VI
SPEED PI CONTROLLER PARAMETERS

Parameter	Equation	Value
$K_{p\omega}$	$\frac{J}{k_b} \frac{2\pi f_{bi}}{a}$	62.0455
$K_{i\omega}$	$K_{p\omega} \frac{2\pi f_{bi}}{a^2}$	48731.9174
<i>Limits</i>	$\frac{60P}{2\pi N \times K_b}$	± 38 A

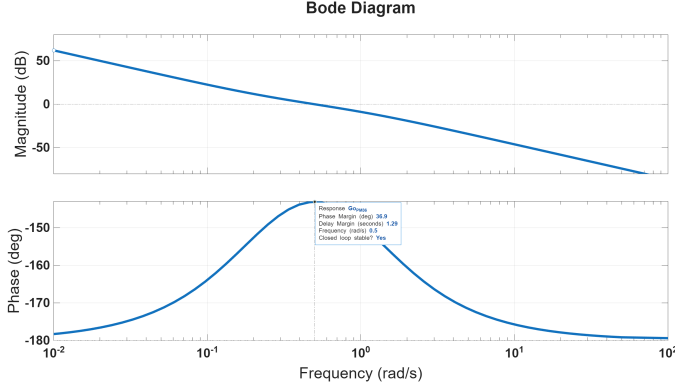


Fig. 9. Bode Plot

For a normalized function ($\tau_{bi} = 1$), the maximum phase margin frequency is 0.5 rad/s .

The speed response, current response, and disturbance rejection characteristics of the system were obtained for a unit step input and are given by Fig. 10, Fig. 11, and Fig. 12 respectively.

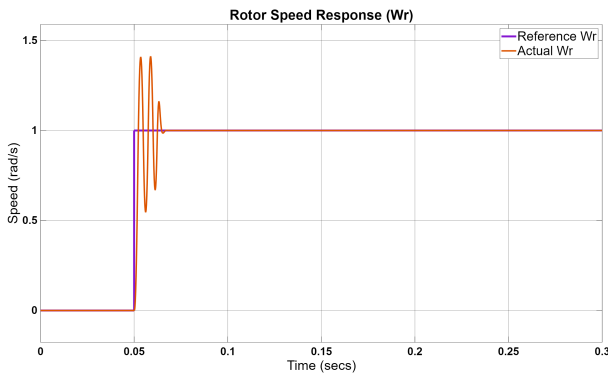


Fig. 10. ω_r v/s t

2) Phase Margin of 55° :

The standard formula for calculating the value of parameter a for any phase margin is given by,

$$PM = 180 + \angle(-180 + \arctan(a) - \arctan(\frac{1}{a}))$$

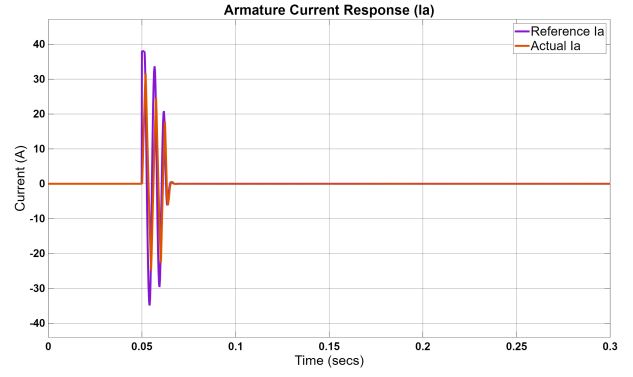


Fig. 11. I_a v/s t

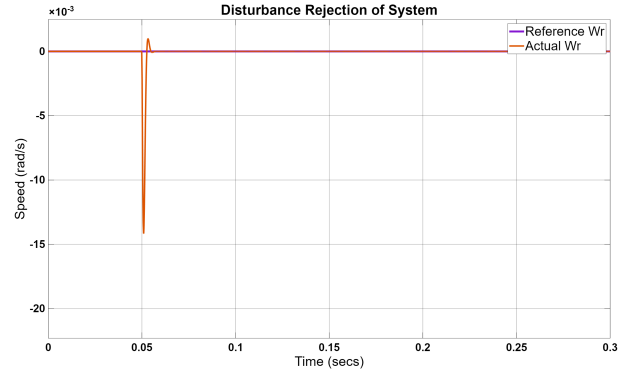


Fig. 12. ω_r v/s t

For a phase margin of 55° , $a = 3.1716$. The open loop transfer function is given by:

$$G_0(s) = \frac{10.06\tau_{bi}s + 1}{31.9\tau_{bi}^3s^3 + 31.9\tau_{bi}^2s^2}$$

The calculated controller parameters are given in the Table VII.

TABLE VII
SPEED PI CONTROLLER PARAMETERS

Parameter	Equation	Value
$K_{p\omega}$	$\frac{J}{k_b} \frac{2\pi f_{bi}}{a}$	39.1226
$K_{i\omega}$	$\frac{J}{k_b} \frac{2\pi f_{bi}}{a^2}$	12218.5496
<i>Limits</i>	$\frac{60P}{2\pi N \times K_b}$	± 38 A

The bode plot of the open loop gain is given in Fig. 13.

It can be observed from Fig. 13 that the maximum phase margin of the system, 55° , occurs at 0.315 rad/s which is again the geometric mean of the two corner frequencies of the open loop function.

The speed response, current response, and disturbance rejection characteristics of the system were obtained for a unit

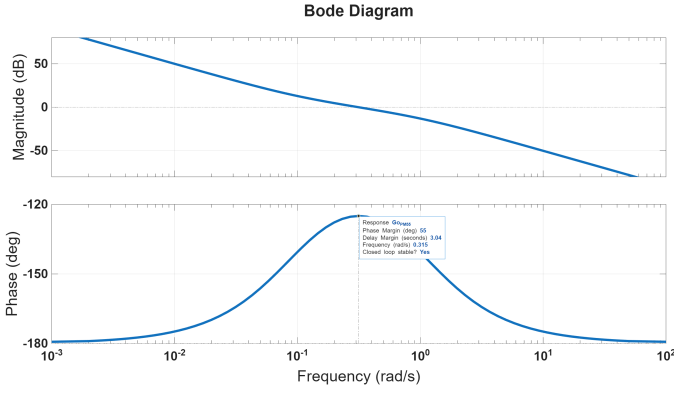


Fig. 13. Bode Plot

step input and are given by Fig. 14, Fig. 15, and Fig. 16 respectively.

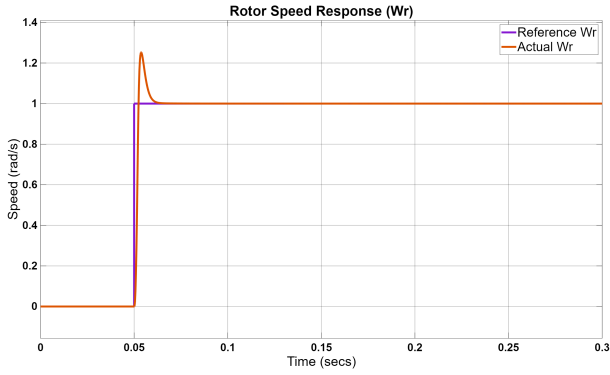


Fig. 14. ω_r v/s t

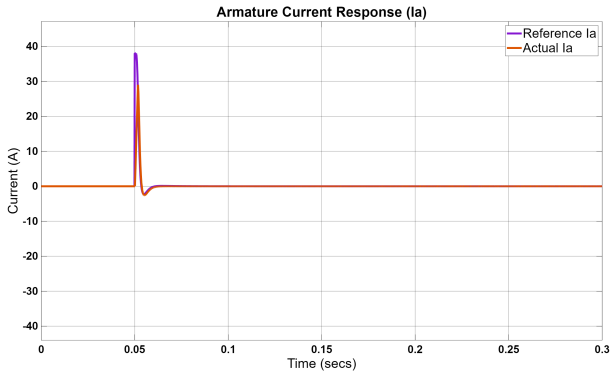


Fig. 15. I_a v/s t

3) Comparison Between the Two Designs:

From Table VIII and Table IX, the differences in the responses for different phase margins can be observed. The system with a phase margin of 55° as compared to the 36.9° phase margin system, exhibits lesser oscillations, faster settling ($t_{s55} < t_{s36}$), increased rise time ($t_{r55} > t_{r36}$), with a smaller

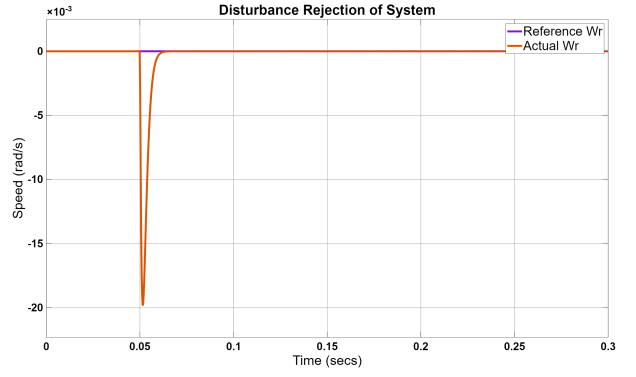


Fig. 16. ω_r v/s t

TABLE VIII
COMPARISON BETWEEN SPEED RESPONSES

PM	36.9°	PM	55°
t_r	1.3 ms	t_r	1.4 ms
t_s (2%)	14.9 ms	t_s (2%)	9.9 ms
%Overshoot	40.91%	%Overshoot	25.24%

TABLE IX
COMPARISON BETWEEN DISTURBANCE REJECTION CHARACTERISTICS

PM	36.9°	PM	55°
t_{fall}	1.1 ms	t_{fall}	4.7 ms
t_s (2%)	4.8 ms	t_s (2%)	13 ms
Undershoot	-14.2×10^{-3} rad/s	Undershoot	-19.8×10^{-3} rad/s

overshoot as well ($Overshoot_{s55} < Overshoot_{s36}$).

The disturbance rejection capabilities have reduced, as the system experiences a greater drop in speed ($Undershoot_{s55} > Undershoot_{s36}$), and requires a longer time to recover and return to the reference value ($t_{fall55} > t_{fall36}$ and $t_{s55} > t_{s36}$). The standard $4\tau_{bi}$ method was designed to allow the system to achieve the maximum feasible bandwidth, which resulted in a phase margin of 36.9° . For any desired phase margin greater than 36.9° , the bandwidth decreases, leading to a slower response. This slower response would reduce oscillations, and leads to a faster settling, however, after a disturbance, the system takes a longer time to recover.

III. INDUCTION MACHINE FLUX CONTROL

In this section, two different control structures for the flux regulation of an induction machine have been analyzed. For each control loop, the steady-state error, transient response, stability margins, and disturbance rejection characteristics have been compared.

A. Model Parameters

The two control systems are designed using the machine parameters given in Table X.

TABLE X
INDUCTION MACHINE PARAMETERS

Parameter	Symbol	Value
Rated Power	P	30 kW
Rated Voltage	V	380 V
Rated Speed	N	1450 rpm
Frequency	f	50 Hz
Stator Resistance	R_s	0.1273Ω
Rotor Resistance (referred)	R'_r	0.1273Ω
Stator Inductance	L_{ls}	1.341 mH
Stator Inductance (referred)	L'_{lr}	1.341 mH
Magnetizing Inductance	L_m	45.219 mH
Moment of Inertia	J	1.6 kgm ²
Total Leakage Factor	σ	0.0568
Rotor Time Constant	τ_r	0.3657 s
Poles	P_N	4

B. Single Loop Control

The flux control loop considered is shown in Fig. 17.

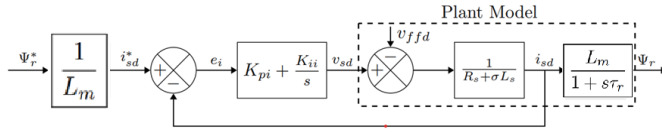


Fig. 17. Single Loop Control

The controller parameters have been calculated for a current loop bandwidth of $f_{bi} = 500 \text{ Hz}$, which are shown in Table XI.

TABLE XI
PI CURRENT CONTROLLER PARAMETERS

Parameter	Equation	Value
K_{pi}	$2\pi f_{bi}\sigma(L_{ls} + L_m)$	8.3083
K_{ii}	$2\pi f_{bi}R_s$	399.9247
Limits	eq. 10	$\pm 104.6937 \text{ V}$

The flux transient response, and disturbance rejection characteristics are obtained for a unit step input and are given in Fig 18 and Fig. 19 respectively.

C. Two Loop Control

The flux control loop considered is shown in Fig. 20.

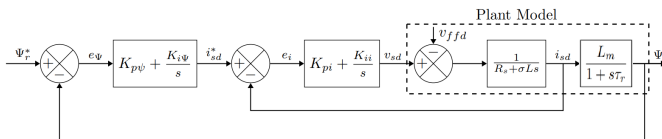


Fig. 20. Double Loop Control

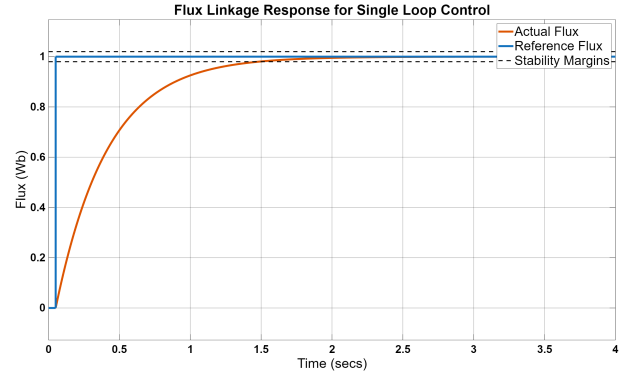


Fig. 18. Ψ v/s t

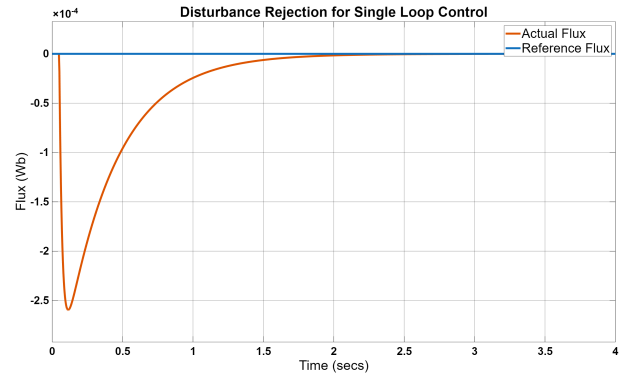


Fig. 19. Ψ v/s t

The controller parameters have been calculated for current loop bandwidth of $f_{bi} = 500 \text{ Hz}$, which are shown in Table XII.

The current controller parameters will be the same as obtained in Table XI, the values in Table XII are the calculated values for the flux controller.

TABLE XII
PI FLUX CONTROLLER PARAMETERS

Parameter	Equation	Value
$f_{b\Psi}$	$\frac{f_{bi}}{10}$	50 Hz
$K_{p\Psi}$	$2\pi f_{b\Psi} \frac{\tau_r}{L_m}$	2541.0516
$K_{i\Psi}$	$2\pi f_{b\Psi} \frac{1}{L_m}$	6947.5058
Limits	eq. 8	$\pm 34.2721 \text{ A}$

The flux transient response, and disturbance rejection characteristics are obtained for a unit step input and are given in Fig 21 and Fig. 22 respectively.

D. Comparison Between the Two Loops

The differences in both loops can be observed from Table XIII and Table XIV. Both systems exhibit a first order response, with the two loop control structure showing

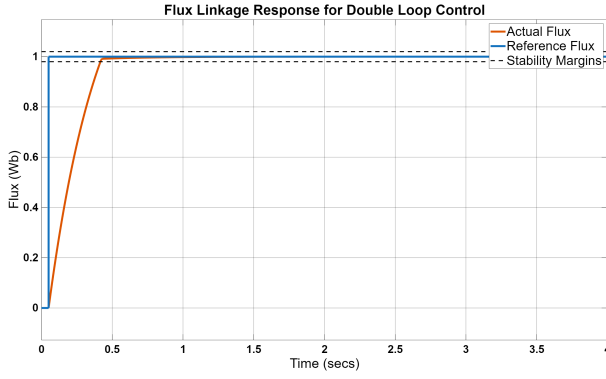


Fig. 21. Ψ v/s t

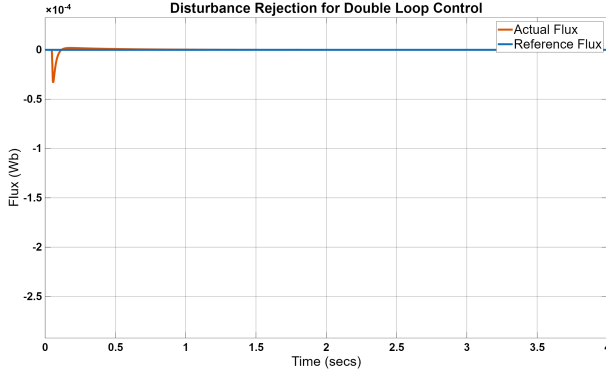


Fig. 22. Ψ v/s t

a superior performance. It is faster ($t_{rTwo} < t_{rOne}$ and $t_{sTwo} < t_{sOne}$) and maintains a highly stable phase margin, in contrast to the marginal stability of the single-loop structure. The two loop structure also provides significant disturbance rejection capabilities ($t_{fallTwo} < t_{fallOne}$, $t_{sTwo} < t_{sOne}$, and $Undershoot_{Two} < Undershoot_{One}$).

E. Calculation of Limits

This section describes the equations involved in the calculation of the controller limits.

$$T_e = \frac{60P}{2\pi N} \quad (5)$$

$$\omega_{slip} = 2\pi f - \frac{2\pi N}{60} \frac{P_N}{2} \quad (6)$$

$$\Psi_{rd} = \sqrt{\frac{R_r T_e}{\omega_{slip}}} \quad (7)$$

$$i_{sd} = \frac{\Psi_{rd}}{L_m} \quad (8)$$

$$i_{sq} = \frac{\omega_{slip} \tau_r \Psi_{rd}}{L_m} \quad (9)$$

$$V_{sd} = R_s i_{sd} + V_{ffd} = R_s i_{sd} - \sigma(L_m + L_{ls})(2\pi f) i_{sq} \quad (10)$$

TABLE XIII
COMPARISON BETWEEN FLUX RESPONSES

Single	Loop	Double	Loop
t_r	0.8099 s	t_r	0.2938 s
t_s (2%)	1.4309 s	t_s (2%)	0.3675 s
SSE	0	SSE	0
PM	1.69°	PM	84.3°

TABLE XIV
COMPARISON BETWEEN DISTURBANCE REJECTION

Single	Loop	Two	Loop
t_{fall}	0.8049 s	t_{fall}	0.0368 s
t_s (2%)	1.7518 s	t_s (2%)	0.2938 s
$Undershoot$	$-26 \times 10^{-5} \text{ Wb}$	$Undershoot$	$-3.3 \times 10^{-5} \text{ Wb}$

IV. CONCLUSIONS

This paper has presented a comprehensive design and analysis of cascaded PI control for a 10 HP DC motor with full bridge converter and flux control of a 30 kW induction motor. Key findings include:

- 1) When a DC motor is operated without a speed controller, and a current input is directly applied, the speed increases unregulated until the maximum output voltage of the converter is reached. Beyond this point, the armature current falls to value determined by the load torque.
- 2) For a current loop bandwidth, $f_{bi} > \frac{f_{sw}}{6}$, the system will be pushed to marginal stability.
- 3) A speed controller designed with a phase margin $> 36.9^\circ$ will experience fewer oscillations and faster settling, but with a slower rise time and worsened disturbance rejection characteristics due to the system's reduced bandwidth.
- 4) When an induction machine is designed with two loop control for flux regulation, it will exhibit a faster response, and improved disturbance rejection characteristics.